GBU 2301: Business Statistics

Practice Questions: Discrete Probability Distributions I and II

Fall 2026 | Dr. Ömer ÇAYIRLI

Without solutions

Choose one answer for each multiple-choice question. For the problems, show calculations and state units. A scientific calculator is sufficient; software is not required. Keep intermediate values unrounded.

For final answers, use exact values where convenient; otherwise, four decimal places for probabilities and two for other numerical results are sufficient.

Each question is independent unless stated otherwise.

Discrete Probability Distributions I

Multiple-Choice Questions

Questions 1–15.

Question 1 (Easy)

Alpha Consulting randomly awards one of four contracts. Let X be the fee earned.

Contract	Beta	Gamma	Delta	Epsilon
Fee (MAD)	12.50	20.00	27.50	50.00

Which statement is correct?

A. X is continuous because some fees contain decimals.

B. X is discrete because it has four possible values.

C. X is continuous because money has a meaningful zero.

D. X is discrete only if the four contracts are equally likely.

Question 2 (Easy)

Beta Repairs records the number of urgent jobs received on 20 comparable days.

Urgent jobs, x	0	1	2	3
Number of days	6	10	3	1

Using these relative frequencies, what is the estimated probability of at least two urgent jobs on a comparable day?

A. 0.05

B. 0.15

C. 0.20

D. 0.80

Question 3 (Average)

Gamma Logistics uses the following distribution for a daily adjustment.

Adjustment, X (hundred MAD)	-1	0	2	4
Probability	k	$2k$	0.25	0.15

These are the only possible adjustments. What is the probability that the adjustment is zero or negative?

A. 0.40

B. 0.20

C. 0.25

D. 0.60

Question 4 (Easy)

Delta Retail offers four equally likely voucher values.

Voucher value (MAD)	0	10	30	40

Which statement correctly describes the expected voucher value?

A. It is 20 MAD, although no individual voucher has that value.

B. It is 20 MAD, so 20 MAD is the most likely voucher value.

C. It is 10 MAD because an expected value must be one of the possible values.

D. It is 40 MAD because repeated draws eventually produce the largest value.

Question 5 (Easy)

For Epsilon Services' daily number of cancellations, $E(X) = 3$ and $E(X^2) = 13$. What are the variance and standard deviation?

A. Variance 10; standard deviation $\sqrt{10}$

B. Variance 4; standard deviation 2

C. Variance 13; standard deviation $\sqrt{13}$

D. Variance 2; standard deviation 4

Question 6 (Average)

Zeta Services starts with a daily allowance of 200 MAD. Each service call uses 30 MAD. If X is the number of calls, the remaining balance is $Y = 200 - 30X$.

Measure of X	Value
Mean	2 calls
Variance	0.64 calls squared

What are the mean and standard deviation of Y?

A. Mean 140 MAD; SD -24 MAD

B. Mean 140 MAD; SD 19.20 MAD

C. Mean 200 MAD; SD 24 MAD

D. Mean 140 MAD; SD 24 MAD

Question 7 (Easy)

Eta Supplies has two pending orders. Let X and Y indicate whether the first and second orders are confirmed: 1 for confirmed, 0 otherwise. The table gives their joint probabilities.

X	$Y = 0$	$Y = 1$
0	0.20	0.10
1	0.30	0.40

What is the probability that exactly one order is confirmed?

A. 0.10

B. 0.30

C. 0.40

D. 0.70

Question 8 (Average)

Theta Transport records whether two vehicles arrive late. Each indicator is 1 for late and 0 for on time. The table gives their joint probabilities.

First vehicle, X	Second: $Y = 0$	Second: $Y = 1$
0	0.10	0.20
1	0.30	0.40

Given that the first vehicle is late, what is the probability that the second is late?

A. 0.4000

B. 0.5714

C. 0.6000

D. 0.6667

Question 9 (Easy)

Iota Sales has two independent customer decisions. The first customer buys with probability 0.60 and the second with probability 0.25.

What is the probability that both customers buy?

A. 0.15

B. 0.25

C. 0.35

D. 0.85

Question 10 (Average)

Kappa Retail's two outlets have daily sales counts X and Y.

Measure	Value
$\operatorname{Var}(X)$	4
$\operatorname{Var}(Y)$	9
$\operatorname{Var}(X+Y)$	7

Which pair is correct?

A. Covariance -6; correlation -1

B. Covariance 3; correlation 0.50

C. Covariance -3; correlation -0.50

D. Covariance 0; correlation 0

Question 11 (Hard)

Lambda Testing uses three possible paired outcomes.

Outcome	Probability	X	Y
A	0.25	-2	4
B	0.50	0	0
C	0.25	2	4

Which statement is correct?

A. X and Y are dependent and have positive covariance.

B. X and Y are independent and have zero covariance.

C. X and Y are dependent and have negative covariance.

D. X and Y are dependent and have zero covariance.

Question 12 (Average)

Mu Investments places half its initial capital in each of two hypothetical funds.

Measure	Fund Alpha	Fund Beta
Return SD (percentage points)	6	4

The correlation of returns is zero. What is the portfolio's return standard deviation?

A. 5.000 percentage points

B. 3.606 percentage points

C. 13.000 percentage points

D. 2.000 percentage points

Question 13 (Easy)

Nu Couriers' contract requires the company to pay an additional fixed daily charge of 40 MAD. The distribution of deliveries and all other revenues and costs remain unchanged.

How does this affect the distribution of daily net profit?

A. Both its mean and standard deviation fall by 40 MAD.

B. Its mean is unchanged, but its standard deviation falls by 40 MAD.

C. Its mean falls by 40 MAD and its variance falls by 40^2.

D. Its mean falls by 40 MAD and its standard deviation is unchanged.

Question 14 (Average)

Xi Services has two probability models for daily demands X and Y. Both models give the same marginal distribution for each demand, but different joint distributions.

Which result must be the same in both models?

A. The expected combined demand, $E(X+Y)$

B. The variance of combined demand, $\mathrm{Var}(X+Y)$

C. The probability of meeting both demands with a shared capacity

D. The correlation between the two demands

Question 15 (Average)

Omicron Investments uses the following one-year return scenarios. Portfolio weights are nonnegative and sum to one.

State	Probability	Alpha return (%)	Beta return (%)
Weak	0.50	0	12
Strong	0.50	12	4

Which allocation gives an expected return of 7%, and what is its return standard deviation?

A. 25% Alpha, 75% Beta; SD 1 percentage point

B. 50% Alpha, 50% Beta; SD 5 percentage points

C. 50% Alpha, 50% Beta; SD 1 percentage point

D. 75% Alpha, 25% Beta; SD 5 percentage points

Discrete Probability Distributions I

Problems

Questions 16–25.

Question 16 (Easy)

Pi Repairs records urgent jobs on 40 comparable days.

Urgent jobs, x	0	1	2	3
Number of days	6	14	12	8

a. Construct the empirical probability distribution. Calculate its mean, variance and standard deviation.

b. If the same distribution applies on each of the next 100 days, how many of those days are expected to have at least two urgent jobs?

c. The team can complete two urgent jobs per day. Additional jobs are outsourced. Find the probability of outsourcing and the expected number outsourced per day.

Question 17 (Average)

Rho Deliveries receives a daily service adjustment.

Adjustment, X (MAD)	-30	0	60	120
Probability	a	0.20	$2a$	0.20

These are the only outcomes.

a. Find a. Calculate the expected adjustment and its standard deviation.

b. Find the probability that the adjustment is below its expected value.

c. A new fixed daily fee $F \geq 0$ will be deducted from the adjustment. Find the greatest fee that leaves a nonnegative net adjustment on at least 60% of days. Breaking even counts as nonnegative.

Question 18 (Average)

Sigma Sales has two pending offers. The following probabilities describe all possible acceptance outcomes.

Client Beta	Client Gamma	Probability
Declines	Declines	0.25
Declines	Accepts	0.15
Accepts	Declines	0.35
Accepts	Accepts	0.25

Financial item	Amount
Contribution from Beta's accepted offer	50 MAD
Contribution from Gamma's accepted offer	80 MAD
Fixed cost, paid regardless of outcomes	20 MAD

a. Let X be the number of accepted offers. Find its distribution.

b. Let Y be net profit. Find its distribution, mean and standard deviation.

c. Find $P(Y \geq 50)$. If exactly one offer is accepted, is net profit determined?

Question 19 (Average)

Tau Retail can choose either payment arrangement below. The choice does not change the sales distribution. The entries are net daily profits after all costs.

Units sold, x	Probability	Arrangement A (MAD)	Arrangement B (MAD)
0	0.25	−40	10
1	0.50	40	50
2	0.25	120	90

a. Calculate each arrangement's expected profit and standard deviation.

b. The owner requires expected profit of at least 45 MAD and a profit standard deviation no greater than 35 MAD. Which arrangement qualifies?

c. Which arrangement gives the greater probability of earning at least 100 MAD in a day? Explain why this comparison need not produce the same choice as part (b).

Question 20 (Average)

The table gives the joint probabilities of order counts at Upsilon Supplies' two outlets on the same day.

First outlet, X	Second outlet: $Y = 0$	Second outlet: $Y = 1$
0	0.10	0.20
1	0.25	0.15
2	0.20	0.10

a. Obtain both marginal distributions. Calculate the mean and variance of each order count.

b. Find $P(Y = 1 \mid X = 2)$. Are the order counts independent?

c. Find the distribution and variance of the total number of orders, $S = X + Y$.

Question 21 (Average)

Phi Couriers uses three possible demand scenarios for two routes.

Scenario	Probability	Route A orders, X	Route B orders, Y
A	0.25	0	2
B	0.50	2	1
C	0.25	4	0

a. Calculate both means and variances, covariance, and correlation.

b. Find the variance of total orders $X + Y$, first from the total-order distribution and then using covariance.

c. Does the correlation imply that total orders are constant? Support your answer numerically.

Question 22 (Average)

Chi Repairs' daily contribution before a two-day fixed expense has the distribution below. Contributions on the two days are independent.

Daily contribution (MAD)	0	40	100
Probability	0.20	0.50	0.30

a. Construct the distribution of total contribution over two days. Show how repeated totals are combined.

b. Calculate the mean and standard deviation of the two-day total.

c. A fixed expense of 90 MAD must be covered by the two-day total. What is the probability of full coverage?

Question 23 (Hard)

Psi Investments divides 10,000 MAD between two hypothetical funds for one year. There are no fees, borrowing or short sales.

State	Probability	Alpha return (%)	Beta return (%)
Weak	0.20	-10	12
Moderate	0.50	10	6
Strong	0.30	20	0

a. Find the allocation giving an expected portfolio return of 7.20%. State both weights and investment amounts.

b. Calculate the funds' variances and covariance. Use them to calculate the portfolio variance and standard deviation.

c. Check the portfolio's mean and variance directly from its three scenario returns. Find the probability of a negative portfolio return under this model.

Use percentage numbers in the calculations: 10 means 10%.

Question 24 (Hard)

Omega Foods stocks one, two or three boxes before daily demand is known.

Demand, D (boxes)	0	1	2	3
Probability	0.10	0.25	0.40	0.25

Item	Amount per box
Purchase cost	25 MAD
Selling price	60 MAD
End-of-day salvage value of an unsold box	5 MAD

Sales cannot exceed stock. Unmet demand is lost, with no additional penalty. There are no other costs, and stocking choice does not affect demand.

Recommend the stocking quantity with the highest expected profit among those that satisfy all demand on at least 90% of days. Compare it with the choice that maximizes expected profit without this service requirement.

Question 25 (Hard)

Beta Packaging has the following joint distribution for two counts recorded on a production run. These are the only possible pairs.

First count, X	Second count, Y	Probability
0	1	0.25
1	0	0.50
2	1	0.25

a. Calculate both means and variances, covariance, and correlation.

b. Determine whether the counts are independent, using a probability comparison.

c. Construct the distribution of $S = X + Y$. Does its variance equal the sum of the two individual variances? Explain how your answer is consistent with part (b).

Discrete Probability Distributions II

Multiple-Choice Questions

Questions 26–40.

Question 26 (Easy)

Gamma Services is choosing a model for a count. In which case is a binomial model exact under the stated conditions?

A. Count customers contacted until three purchases occur.

B. Count certified employees among four people randomly selected without replacement from a team of ten.

C. Count purchases among six customers who decide independently, each with purchase probability 0.30.

D. Count purchases among six independent customers, three with purchase probability 0.20 and three with probability 0.60.

Question 27 (Easy)

Delta Components tests four independently produced items. Each is defective with probability 0.30.

What is the probability that exactly one is defective?

A. 0.4116

B. 0.1029

C. 0.6517

D. 0.0081

Question 28 (Average)

Epsilon Couriers pays a 30-MAD refund if a parcel is late and nothing otherwise. The parcel is late with probability 0.20.

What are the mean and standard deviation of the refund?

A. Mean 6 MAD; SD 4.80 MAD

B. Mean 24 MAD; SD 12 MAD

C. Mean 6 MAD; SD 24 MAD

D. Mean 6 MAD; SD 12 MAD

Question 29 (Average)

Zeta Sales uses a binomial model for the number of sales in a campaign. Its theoretical mean is 3 and its theoretical variance is 2.25.

Which parameters produce these moments?

A. $n = 4,\ p = 0.75$

B. $n = 12,\ p = 0.25$

C. $n = 9,\ p = 1/3$

D. $n = 3,\ p = 0.75$

Question 30 (Easy)

Eta Sales models the number of purchases as $X \sim \mathrm{Bin}(6, 0.40)$. The cumulative probabilities are:

Event	Probability
$X \le 1$	0.23328
$X \le 2$	0.54432

What is the probability of more than two purchases?

A. 0.54432

B. 0.76672

C. 0.45568

D. 0.23328

Question 31 (Easy)

Theta Support receives calls according to a Poisson process at four calls per hour. What is the probability of no calls in 15 minutes?

A. 0.0183

B. 0.3679

C. 0.6321

D. 0.7788

Question 32 (Average)

Iota Dispatch receives requests according to a constant-rate Poisson process. The expected count in 90 minutes is six requests.

What is the standard deviation of the count in 30 minutes?

A. 2 requests

B. $\sqrt{6}$ requests

C. 3 requests

D. $\sqrt{2}$ requests

Question 33 (Average)

Kappa Reservations receives calls according to a Poisson process. Each ten-minute interval has expected count 1.5.

What is the probability that each of two consecutive ten-minute intervals contains at least one call?

A. 0.6035

B. 0.7769

C. 0.9502

D. 0.0498

Question 34 (Average)

Lambda Cables uses a Poisson model with mean two flaws per inspected length. A length is rejected if it has at least three flaws.

What is the rejection probability?

A. 0.1804

B. 0.6767

C. 0.3233

D. 0.5940

Question 35 (Average)

Mu Support's 90 half-hour observations have sample mean 4 calls and sample variance 11. Calls also appear more frequent around lunchtime.

Which conclusion is best supported?

A. A Poisson model is impossible because sample variance must exactly equal sample mean.

B. A single Poisson model with mean 11 resolves the difference between the summaries.

C. The mean of 4 is sufficient to establish a constant-rate Poisson model.

D. Variation in the rate may contribute to the larger observed variance; the constant-rate assumption needs examination.

Question 36 (Easy)

Nu Services randomly selects five different employees from a team of nine. Six team members are certified.

What are the possible counts of certified employees in the selected group?

A. 2, 3, 4, 5

B. 0, 1, 2, 3, 4, 5

C. 0, 1, 2, 3

D. 1, 2, 3, 4, 5, 6

Question 37 (Average)

Xi Imports randomly selects two different items from a lot of eight, exactly three of which are defective. Every pair is equally likely.

What is the probability of exactly one defective item in the pair?

A. 0.3750

B. 0.5357

C. 0.4688

D. 0.6429

Question 38 (Average)

Omicron Testing draws four items from 20, exactly five of which are defective.

Compare the variance of the defect count under random sampling without replacement with the variance under independent draws with replacement. What is the ratio

$$\frac{\text{variance without replacement}}{\text{variance with replacement}}?$$

A. 1

B. 19/16

C. 4/20

D. 16/19

Question 39 (Average)

Pi Components draws items sequentially without replacement from a lot of 12 containing exactly four defective items. Each draw selects uniformly at random from the remaining items.

The first item drawn is nondefective. What is the probability that both of the next two items are defective?

A. $(4/12)(3/11)$

B. $(4/11)^2$

C. $(4/11)(3/10)$

D. $(3/11)(2/10)$

Question 40 (Average)

Rho Testing compares random samples without replacement. Each population has a 10% defect proportion.

Plan	Population size, N	Defective items, r	Sample size, n
A	10,000	1,000	5,000
B	200	20	4
C	1,000	100	100
D	500	50	100

Based on the sample fraction, which plan gives the strongest justification for a binomial approximation?

A. Plan A

B. Plan B

C. Plan C

D. Plan D

Discrete Probability Distributions II

Problems

Questions 41–50.

Question 41 (Easy)

Sigma Transport sends five parcels. On-time delivery outcomes are independent, with probability 0.70 for each parcel.

a. Find the probability that exactly three parcels arrive on time and the probability that at least four arrive on time.

b. Calculate the mean, variance and standard deviation of the number arriving on time.

Question 42 (Average)

Tau Components tests five independently produced items. Each is defective with probability 0.20. A batch is sent for additional processing if at least two items are defective.

Cost	Amount
Routine inspection	10 MAD per batch, always paid
Repair of a defective item	60 MAD per defective item
Additional processing	100 MAD per batch when at least two items are defective

The additional processing charge is paid once per qualifying batch, on top of inspection and repair costs. There are no other costs.

a. Find the probability that additional processing is required.

b. Given that additional processing is required, find the probability of exactly two defective items.

c. Calculate expected total cost per batch.

Question 43 (Average)

Upsilon Sales contacts two customers whose decisions are independent.

Customer	Purchase probability
Beta	0.20
Gamma	0.60

Let X count purchases.

a. Construct the complete distribution of X. Calculate its mean and variance.

b. Compare your distribution with a binomial model having $n = 2$ and $p = 0.40$. Calculate its probabilities, mean and variance. Is this an exact replacement for the actual model?

Question 44 (Easy)

Phi Support receives calls according to a constant-rate Poisson process.

Arrival rate	Interval of interest
9 calls per hour	20 minutes

a. State the distribution of the interval count and calculate its mean, variance and standard deviation.

b. Find the probability of exactly two calls and the probability of at most two calls.

Question 45 (Average)

Chi Printing models flaws by a Poisson process with a constant rate of 0.20 flaws per square metre. Counts on separate sheets are independent.

Three sheets are inspected. Each has area four square metres. A sheet passes if it has at most one flaw.

a. Find the probability that one sheet passes.

b. Find the probability that at least two of the three sheets pass.

c. Compare the probability that all three sheets pass with the probability that there is at most one flaw across the three sheets in total. Explain why the events differ.

Question 46 (Hard)

Psi Reservations has one operating condition throughout a randomly selected 20-minute window. The condition is initially unknown. Conditional on that condition, arrivals follow a Poisson process at the stated rate.

Condition	Probability	Arrival rate
Regular	0.60	3 calls per hour
Busy	0.40	12 calls per hour

a. Find the probability of exactly two calls in the window.

b. The window contains exactly two calls. Find the probability that the condition was busy. Does this evidence increase or decrease the busy-condition probability?

c. A single Poisson model is fitted using the probability-weighted average rate. Calculate its probability of exactly two calls and compare it with part (a).

Question 47 (Average)

Omega Services selects three different employees from a team of nine. Four are certified. Every group of three is equally likely.

a. Construct the complete probability distribution of the number of certified employees selected.

b. Calculate its mean, variance and standard deviation.

c. Find the probability that the group contains both certified and non-certified employees.

Question 48 (Average)

Beta Testing compares two random sampling procedures for a lot of eight items, exactly three of which are defective.

Procedure	Sampling rule
A	Draw four different items without replacement.
B	Draw four times, replacing the selected item and mixing the lot after every draw.

For B, draws are independent. A defective item may be selected more than once and counts on each draw.

a. State the exact model and possible defect counts for each procedure.

b. Calculate the mean and variance under each procedure.

c. Find the probability of at least three recorded defects under each procedure. Is the binomial model a good substitute for A for this event?

Question 49 (Hard)

Gamma Imports receives a lot of ten items containing exactly three defective items. Tests identify defects perfectly.

The inspection rule is:

- Randomly select and inspect two different items.
- If either is defective, reject the lot and stop.
- Otherwise, randomly select and inspect two more items from those remaining.
- Accept the lot only if all four inspected items are nondefective.

Each item inspected costs 3 MAD. Both items in a stage are tested before the decision for that stage. There are no other inspection costs.

a. Find the probability of reaching the second stage and the probability of eventual acceptance.

b. Calculate the expected number of items inspected and expected inspection cost.

c. Compare this rule with always inspecting four different items and accepting only if all four are nondefective. Does stopping early change the acceptance probability or expected cost?

Question 50 (Hard)

Delta Rentals randomly selects three different devices from eight in storage. Exactly two of the eight are faulty. Every group of three is equally likely. Faulty devices are identified after selection and cannot be rented out.

Three customers each decide independently whether to rent one device, with rental probability 0.50. Customer decisions are independent of the device selection.

Before the selection and customer decisions, Delta must choose whether to reserve one additional device that is guaranteed to work.

Financial item	Amount
Contribution per device rented, after rental-related costs	40 MAD
Fixed reservation fee for the additional device	8 MAD, paid even if unused
Other relevant costs	0 MAD

Each available device can serve one customer. Unmet demand is lost, with no further charge.

a. Find the distributions of usable devices in the original three-device selection and customer demand.

b. For each reservation choice, calculate expected profit and the probability of meeting all customer demand. Recommend the higher-expected-profit option among those meeting all demand with probability at least 95%.